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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution Affiliated to VTU)
IV Semester B. E. Examinations Oct-2023
Common to EC / EI
SIGNALS & SYSTEMS

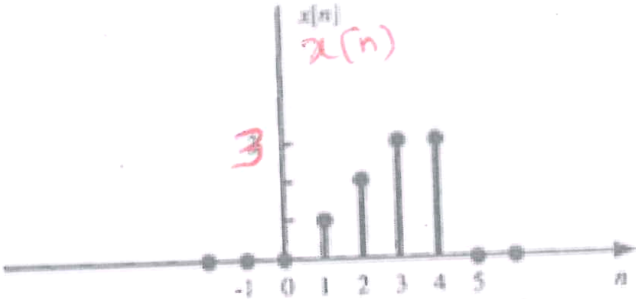
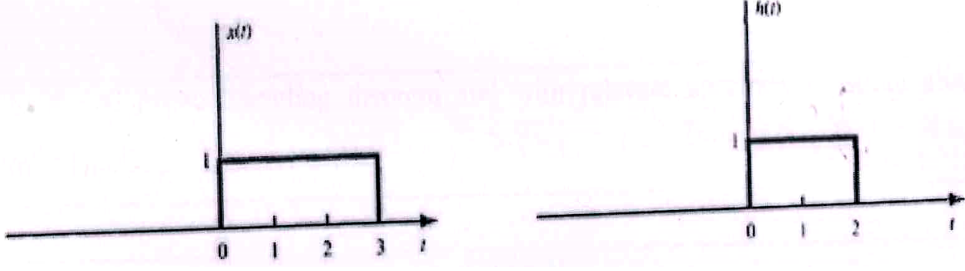
Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

PART-A

1	1.1	Determine the 2 – point DFT of the given sequence $x[n] = [2, 5]$.	01
	1.2	In an N-point DFT, computation of each point of DFT can be accomplished by how many number of complex multiplications and complex additions.	01
	1.3	Consider the data set $x = [2, 3, 5, 7, 11, 13, 17, 19, 23, 29]$, find the interquartile range (IQR).	01
	1.4	Compute the IDFT of the four-point sequence: $X(k) = (6, -2 + 2j, -2, -2 - 2j)$	01
	1.5	State the two properties of phase (twiddle) factor W_N that reduces the number of computations in FFT algorithms.	01
	1.6	Find the relation between correlation and convolution.	01
	1.7	Determine the power and energy of unit step sequence.	01
	1.8	Find the relation between signum function and unit step function.	01
	1.9	Find the fourier transform of $x[n] = -a^n u[-n - 1]$ a is real number.	01
	1.10	What is aliasing in the process of sampling?	01
	1.11	Find the correlation of the symmetrical gate pulse of amplitude and time duration of '1' with itself.	02
	1.12	Consider the analog signal $x(t) = \sin 200\pi t . \sin 1000\pi t$. Determine the fundamental period.	02
	1.13	Determine the even and odd components of the signal: $x[n] = \delta[n]$.	02
	1.14	Consider a discrete-time LTI system with impulse response $h[n] = \begin{cases} 1, & n = 0, 1 \\ 0, & \text{otherwise} \end{cases}$ Find the input-output relationship of the system.	02
	1.15	Determine whether or not each of the following signals is periodic. If a signal is periodic, determine its fundamental period. $x[n] = \cos \frac{\pi}{3}n + \sin \frac{\pi}{4}n$	02

PART-B

2 a i) $Y[n] = \sum_{k=-\infty}^{n+1} x(k)$ ii) $y[n] = x[n] \cos(\omega_0 n)$ b i) $x[n-2]$ ii) $x[2n]$ iii) $x[-n]$ iv) $x[-n+2]$	With explanation determine if the given systems are static, linear, time invariant, non-causal and unstable. A discrete-time signal $x(n)$ is shown in Fig. 2.b. Sketch and label each of the following signals:	08
	 Fig. 2.b	05
	c Categorize the given signal as power or energy signal. Also determine the power and energy of the signal.	03
3 a i) Formulate the operator for the system relating $y[n]$ to $x[n]$ ii) Develop the block diagram representation for cascade and parallel implementation. b i) Analytical method ii) Graphical method	The output of a discrete-time system is related to its input $x[n]$ as follows: $y[n] = x[n] + x[n-2] + x[n-3] + x[n-4]$ The input $x[n]$ and the impulse response $h[n]$ of a discrete-time LTI system are given by $x[n] = u[n]$ and $h[n] = a^n u[n]$. Find convolution sum using	06 10
OR		
4 a i) Find the impulse response $h[n]$ of the system ii) Consider the impulse response $h[n]$ from above i), is the system causal and stable? b	The step response $s[n]$ of a discrete-time LTI system is given by $s[n] = a^n u(n) \quad 0 < a < 1$ Evaluate $y(t) = x(t) * h(t)$, where $x(t)$ and $h(t)$ are shown in Fig. 4.b  Fig. 4.b	06 10

5	a Consider a system with output $y(t) = g(t)x(t)$. Assume $x(t)$ is the square wave as shown in Fig. 5.a i) Find $Y(j\omega)$ in terms of $G(j\omega)$ ii) Sketch $Y(j\omega)$ if $g(t) = \cos\left(\frac{t}{2}\right)$ Fig. 5.a	08
	b Determine the discrete Fourier series representation for the given signal $x[n]$ i) $x[n] = \cos\frac{\pi}{3}n + \sin\frac{\pi}{4}n$ ii) $x[n] = \cos^2\left(\frac{\pi}{8}\right)n$ Sketch the spectrum for each case. OR	08
6	a The block diagram of a system, consisting of two sub-blocks: one LTI system with impulse response $h(t)$ and a multiplier, is shown in Fig. 6.a. Using Fourier Transform, determine response $y(t)$ of the system for given $x(t) = \cos(\pi t)$ and $g(t) = \cos(3\pi t)$ Fig. 6.a b Find the DTFT of $x[n] = \cos\left(\frac{\pi n}{4}\right) \cdot \cos\left(\frac{\pi n}{2}\right)$. Show all the intermediate steps and spectrums.	08
7	a For the given sequence $x_1(n) = (1, 2, 3, 1)$ and $x_2(n) = (4, 3, 2, 2)$ use the four point DFT and IDFT to determine the sequence $x_3(n)$ corresponding to the circular convolution of the sequences $x_1(n)$ and $x_2(n)$. b For the sequence $X_1(n) = \cos\left(\frac{2\pi}{N}\right)n$, $X_2(n) = \sin\left(\frac{2\pi}{N}\right)n$ $0 \leq n \leq N - 1$ Determine the N point i) Circular convolution of the sequences $x_1(n)$ and $x_2(n)$ ii) Circular autocorrelation of the sequences $x_1(n)$ OR	08
8	a Compute the output of a digital filter whose impulse response is $h[n] = [1, 2, 1]$ for the input signal $x[n] = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]$ using over-lap-add method. Use (i) N-point circular convolution (ii) 8 point circular convolution. b Find the 4 point circular convolution of sequence $x[n] = [1, 1, 1, 1]$ & $h[n] = [1, 0, 1, 0]$ using radix-2 DIF-FFT algorithm.	08

9	a	Using one real time application show the importance of features. Mentions min six features to support the answer.	08
	b	Write short note on the following: i) Mel-Frequency Cepstral Coefficient (MFCC) ii) Short time Fourier Transform (STFT) iii) Chroma	08
	OR		
10	a	Mention the importance of six supporting libraries for the feature extraction of audio signal in python.	08
	b	With neat explanation, mention any six dominant features to distinguish between two audio signals.	08